

Journal Target Recommendation: '02' DGP Architecture Comparison

Contents

Manuscript summary	3
Methodological positioning	3
Most similar papers in the Zotero nof1 collection	3
Ranked journal targets	4
Rank 1: <i>Statistics in Medicine</i>	4
Rank 2: <i>Biometrics</i>	4
Rank 3: <i>Trials</i>	5
Rank 4: <i>Statistical Methods in Medical Research</i>	5
Rank 5: <i>Contemporary Clinical Trials Communications</i>	5
Rank 6: <i>Harvard Data Science Review</i>	5
Rank 7: <i>Healthcare</i> (MDPI)	6
Primary recommendation	6

List of Tables

1	Most similar papers in the Zotero nof1 collection with similarity and journal	3
---	--	---

Manuscript summary

‘Two Architectures for Simulating Biomarker-Treatment Interactions: Implications for Statistical Power Under Carryover’ formalizes the distinction between direct mean moderation (Architecture A) and differential correlation via MVN (Architecture B) as data-generating processes for biomarker-treatment interaction simulations. The central finding is that the choice of DGP architecture determines whether carryover effects reduce statistical power modestly (~10-15%, Architecture A) or substantially (~40-60%, Architecture B). The manuscript provides mathematical formalization, biological rationale for each architecture, simulation evidence from N-of-1 trial designs, and guidance for DGP selection.

Methodological positioning

The manuscript sits at the intersection of three domains:

- Statistical simulation methodology (DGP specification and its consequences)
- N-of-1 and crossover trial design (carryover, power analysis)
- Biomarker-treatment interaction modeling (predictive biomarker validation)

Most similar papers in the Zotero nof1 collection

Table 1: Most similar papers in the Zotero nof1 collection with similarity and journal

Paper	Similarity	Journal
Yap, Wang & Harhay (2024) – carryover and power in crossover	Highest	<i>Biometrics</i>
Blackston et al. (2019) – simulation comparison, aggregated N-of-1 vs. RCT	High	<i>Healthcare</i>
Wang & Schork (2019) – power and design in crossover N-of-1	High	<i>Healthcare</i>
Senn (2016) – variance components and personalized medicine	High	<i>Statistics in Medicine</i>

Paper	Similarity	Journal
Schork (2022) – serial correlation in personalized studies	High	<i>Harvard Data Science Review</i>
Haller & Ulm (2018) – biomarker-treatment interaction estimation	High	<i>Trials</i>
Arends et al. (2019) – sample size for N-of-1	Moderate	<i>Statistical Methods in Medical Research</i>
Tang & Landes (2020) – t-tests for N-of-1 with serial correlation	Moderate	<i>PLoS ONE</i>
Chen et al. (2024) – methodological review of N-of-1	Moderate	<i>Trials</i>

Ranked journal targets

Rank 1: *Statistics in Medicine*

- **Collection precedent:** Senn (2016)
- **Rationale:** Premier journal for statistical methodology in clinical trial design. The core contribution – that DGP architecture choice produces 40-60% divergent power estimates under carryover – is a methodological finding with implications for how the field designs and reports simulation studies. The journal’s readership (biostatisticians designing trials) is the primary audience who would act on this finding. The Path B strengthened version (1,000+ replicates, evaluated alternative analysis strategies, power curves) meets the journal’s expectations for rigor.
- **Risk:** Competitive acceptance rate. Must demonstrate the finding generalizes beyond the specific pmsimstats parameterization.

Rank 2: *Biometrics*

- **Collection precedent:** Yap, Wang & Harhay (2024)
- **Rationale:** Yap et al. (2024) is the closest comparator paper in the collection – carryover’s impact on power in crossover designs, published in *Biometrics* in 2024. The ‘02’ manuscript extends this by showing that the mechanism by which the biomarker-treatment interaction is generated determines the magnitude of power loss. Natural companion to Yap et al.
- **Risk:** Expects stronger theoretical results (formal proofs, asymptotic analysis). May view the scope as too narrow for a generalist biometrics

readership. Would need broader framing in terms of covariance-vs-mean signal detection.

Rank 3: *Trials*

- **Collection precedent:** Haller & Ulm (2018), Chen et al. (2024), Senn & Julious (2024), Mitchell et al. (2024)
- **Rationale:** Highest concentration of N-of-1 methodology papers in the collection (4 of 50). Haller & Ulm (2018) is essentially the same methodological genre as ‘02’ (simulation study of biomarker-treatment interaction estimation). Open access, practical orientation, readership includes both methodologists and trialists.
- **Risk:** Lower impact factor. Theoretical depth may exceed the journal’s typical level, though this is minor.

Rank 4: *Statistical Methods in Medical Research*

- **Collection precedent:** Arends et al. (2019)
- **Rationale:** Publishes simulation-based methodological work for clinical trial design. Arends et al. (2019) is the same genre (power/sample size for N-of-1). Occupies middle ground between theoretical focus of *Biometrics* and practical focus of *Trials*. Accepts longer methodological papers with extensive simulation.
- **Risk:** Slower review times. Less visibility in the N-of-1 community.

Rank 5: *Contemporary Clinical Trials Communications*

- **Collection precedent:** Sturdevant & Lumley (2021), Liu et al. (2021)
- **Rationale:** Two relevant papers on carryover testing and crossover design. Good fallback. Practical framing and clinical application (prazosin-PTSD) fit well.
- **Risk:** Lower visibility and impact than journals above.

Rank 6: *Harvard Data Science Review*

- **Collection precedent:** Schork (2022)
- **Rationale:** Schork (2022) on related statistical machinery was published here. HDSR publishes methodological perspectives.
- **Risk:** More of a review/perspective venue. Heavy simulation results in the Path B version may not fit format expectations.

Rank 7: *Healthcare* (MDPI)

- **Collection precedent:** Blackston et al. (2019), Wang & Schork (2019)
- **Rationale:** Two of the most similar simulation-comparison papers were published here. Fast review, open access.
- **Risk:** MDPI stigma. If strengthened to Path B standards, the manuscript deserves a higher-impact venue.

Primary recommendation

Target: *Statistics in Medicine*. The manuscript's central claim is about simulation methodology itself. *Statistics in Medicine* reaches the audience most likely to change their practice.

Backup: *Trials*. If *Statistics in Medicine* considers the scope too specialized, *Trials* has the most active N-of-1 methodology pipeline and strong fit with collection precedents.

Generated 2026-04-08. Based on Zotero nof1 collection (50 items) and full reading of docs/02-dgp-mean-moderation-vs-mvn.tex.